

MONITORING, EVALUATION AND LEARNING

Endline Outcome Analysis

Sales revenue, household savings and progress along with the theory of change for refugee and host-community entrepreneurs

Inkomoko MEL | Analysis report and dashboard commentary

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837

clients surveyed across 3 locations

610

existing businesses analyzed

+33.6%

growth in mean monthly sales

75%

of households saving

How to use this report

Sections 1 and 2 give the answer in two pages. Sections 4 and 5 explain how the numbers were produced. Sections 6 to 8 present the results in detail. Sections 9 to 11 turn the results into recommendations and link them to the theory of change.

Every figure traces to a live formula in the companion workbook (Inkomoko_MEL_Analysis_Workbook.xlsx); sheet names are cited in table and figure notes.

Companion files: Inkomoko_MEL_Analysis_Workbook.xlsx, Inkomoko_Cleaned_Dataset.xlsx, Inkomoko_MEL_Presentation.pptx

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1. Executive summary

This report analyses the 2024 endline survey of 837 Inkomoko clients in three locations: Hekima Town and the refugee camps of Lala and Sawa. It asks whether supported entrepreneurs show progress in monthly sales revenue and household savings, who is progressing fastest, and whether the pattern is consistent with Inkomoko's theory of change. All amounts are in Central African CFA francs (CFA).

+9,011 CFA mean net change in monthly sales (95% CI 8,379 to 9,642)	+33.6% growth on mean baseline sales of 26,792 CFA	99% of 610 businesses increased sales (6 declined)	75% of households save; median saver holds 10,000 CFA
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Key findings

- Sales grew almost universally:** Mean monthly sales rose from 26,792 to 35,803 CFA; 99% of businesses gained, with results highly significant from zero (paired $t = 27.98$, $p < 0.001$).
- Absolute gains follow business size; growth rates do not:** Absolute gains scale with firm size (3,203 → 17,091 CFA), but growth rates remain flat at ~33%.
- Location gap:** Sawa Camp businesses gained much more (12,505 CFA vs. 5,692 in Lala), reflecting stronger starting sales and far better access to funds (60% vs. 13%).
- Demographics:** Gender, refugee status, and youth show minimal impact once location is controlled. Women and men differ by <700 CFA, while youth gains (7,930 vs. 9,998 CFA) are lower mainly because most youth are concentrated in low-base Lala Camp.
- Support tiers:** Higher support levels yield larger absolute gains (6,390 → 13,907 CFA), but relative growth is nearly identical (~32–34%). Because the investment tier began with roughly double the baseline sales, the data doesn't yet show advisory or investment raising growth rates.
- Saving is widespread but uneven:** Refugees save more than hosts (87% vs. 66%). Rates are lowest among hosts in Sawa (48%) and women in Hekima (58%), while Lala Camp savers hold the least (9,034 CFA vs. 16,145 in Sawa).
- Theory of change:** Support improved sales and savings, but household resilience cannot yet be assessed due to missing consumption, asset or poverty measures and no baseline savings data.

Priority recommendations

- Design a Lala Camp growth package** focused on market access, working capital and savings groups, because the gap there is about business scale and access to finance rather than growth momentum.
- Evaluate the advisory and investment tiers properly** with a phased roll-out or matched comparison before scaling, since the current data cannot separate programme effect from the larger starting size of investment clients.
- Strengthen business management in Hekima Town**, where only 7% keep records and 7% employ staff, satisfaction is lowest (6.5 of 10), and all six businesses with declining sales are located.
- Close the measurement gaps:** follow up idea-stage clients and closures (all 41 existing businesses lost from the sales analysis were Training-only clients), and add baseline savings, employment and a household consumption module.

Bottom line

Inkomoko clients are growing their businesses and building savings across every segment examined. The next gains for the programme lie in closing the capital and scale gap in Lala Camp, proving the added value of the advisory and investment tiers, and measuring household-level outcomes so that the final link in the theory of change, moving households out of poverty, can be tested.

2. Dashboard at a glance

The Dashboard sheet of the workbook brings the four core views together: net sales change, baseline versus endline sales, the share of households saving, and mean savings of savers, each for the twelve subgroups used throughout this report. The charts below reproduce that dashboard, and the headline indicators beneath them come from the Summary sheet.

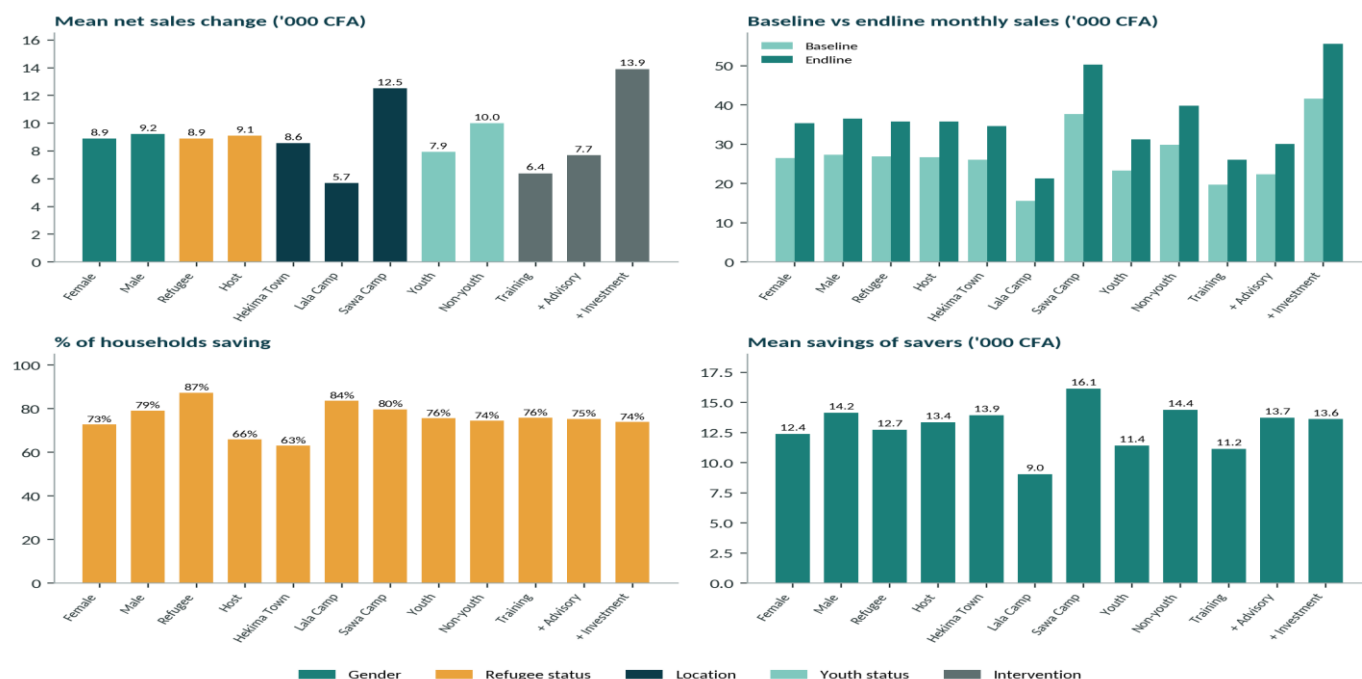


Figure 1. Sales and savings by subgroup. The dimension (gender, refugee status, location, youth status, intervention).

Table 1. Headline indicators (Summary sheet)

Indicator	Value	Definition
Clients surveyed	837	All records in the raw file
Net-sales analysis sample	610	Existing businesses with valid base & end-line sales
Mean baseline monthly sales	26,792 CFA	sales_baseline_monthly_CFA
Mean endline monthly sales	35,803 CFA	sales_endline_monthly_CFA
Mean net change in monthly sales	9,011 CFA	Endline minus baseline
Median net change	6,918 CFA	The typical business (distribution is right-skewed)
Relative growth in mean sales	33.6%	Mean net change divided by mean baseline
Businesses with higher sales	99.0% (604 of 610)	Net change greater than zero
Households saving	75.1% (458 of 610)	liv_saving = yes
Mean savings, savers	13,043 CFA	liv_save_hwmuch_CFA; median 10,000 CFA
Mean savings, all households	9,793 CFA	Non-savers counted as 0 CFA
Correlation: savings and net sales change	0.30	Pearson r among savers

Reading the dashboard. Three patterns stand out. First, the bars for net sales change differ most across locations and support tiers and least across gender and refugee status. Second, the baseline and endline bars rise in near-constant proportion, which is why growth rates look similar even where CFA gains differ. Third, saving rates vary most between refugees and hosts, while the amount saved varies most by location.

3. Background and objectives

Inkomoko and its theory of change

Inkomoko works with refugee and host-community micro and small enterprises (MSEs) to support their growth and resilience. Its theory of change holds that training, advisory services and affordable loans strengthen and expand businesses, which in turn raises sales revenue, creates employment and increases household savings. Together these improve household consumption and self-reliance and, over time, help households move out of poverty.

The endline survey

The dataset is an imaginary 2024 endline survey of 837 entrepreneur clients with particular emphasis on women and youth. It covers three locations (one town and two refugee camps) and records outcomes of interest alongside demographic and programme variables. Missing data is coded inconsistently as 998, 999 or left blank, so identifying and handling missing data is part of the task.

Table 2. *Composition of the 837 surveyed clients*

Dimension	Group	Clients	Share
Location	Sawa Camp	383	45.8%
	Hekima Town	256	30.6%
	Lala Camp	198	23.7%
Gender	Female	529	63.2%
	Male	308	36.8%
Refugee status	Refugee (camp based)	383	45.8%
	Host / non-refugee	454	54.2%
Age (832 with valid age)	Youth (35 or under)	445	53.5%
	Non-youth	387	46.5%
Intervention	Training	380	45.4%
	Training & advisory	296	35.4%
	Training, advisory, & investment	161	19.2%
Business status	Existing business	651	77.8%
	Idea stage	186	22.2%

Objectives and analytical questions

The assignment has three parts. Each is answered in a specific part of this report.

Part	Question	Where answered
A. Data cleaning	How are inconsistently coded missing values identified and handled, what quality checks are run, and what is the cleaning approach?	Section 5; Appendices A to C; cleaned data
B. Outcome analysis	To what extent do supported entrepreneurs show progress in sales revenue and household savings, and how does this differ by gender, refugee status, location, youth status and intervention? What follows for the programmes team?	Sections 2, 6 to 10
C. Theory of change	Do the results show progress along the pathway from business support to expanded businesses, higher sales and savings, and household resilience?	Section 11

4. Methodology and analytical approach

Tools and design

All analysis was carried out in Microsoft Excel using live formulas so that every figure can be audited. The raw data is preserved untouched on the Raw_Data sheet. The Clean_Data sheet rebuilds each variable from the raw data through formulas driven by editable thresholds on the Parameters sheet, and the analysis sheets use COUNTIFS, AVERAGEIFS, SUMIFS, QUARTILE, CORREL and TDIST on the cleaned data. Changing a threshold, such as the youth age cut-off, therefore updates every downstream result.

Key definitions

Term	Definition
Net sales change	sales_endline_monthly_CFA minus sales_baseline_monthly_CFA, for businesses with both values valid.
Relative growth	Mean net change divided by mean baseline sales. It isolates growth from starting size.
Household savings	liv_save_hwmuch_CFA, the amount a saving household reports holding. Measured at endline only, so subgroups are compared as levels, not change.
Youth status	Youth if age is 35 or under; non-youth if over 35. Created from the cleaned age variable.
Refugee status	Refugee if camp-based refugee; host / non-refugee for host-community and secondary-city clients.
Intervention	The support tier received: Training; Training & advisory; or Training, advisory, & investment.
Analysis sample	Existing businesses with valid baseline and endline monthly sales (n = 610).

Statistical approach

- **Subgroup summaries.** For each of the twelve subgroups (and all clients) the workbook reports n, mean baseline and endline sales, mean net change, relative growth, the share of businesses with higher sales, the standard deviation of net change and a 95% confidence interval ($\text{mean} \pm 1.96 \times \text{SD} / \sqrt{n}$).
- **Overall test.** A paired t-test of endline against baseline sales ($t = \text{mean} / (\text{SD} / \sqrt{n})$, with p from TDIST) tests whether average change differs from zero.
- **Two-way splits.** Cross-tabulations (intervention by location, gender by location, refugee status by location, youth by location and intervention) separate location effects from programme effects. Cells with fewer than 30 observations are flagged.
- **Outliers.** Tukey fences ($Q3 + 3 \times \text{IQR}$ and $Q1 - 3 \times \text{IQR}$) flag extreme net changes and savings. Flagged values are retained and a sensitivity analysis shows their influence.
- **Correlation.** Pearson correlations relate baseline sales, percentage growth, net change and savings.

Confidence intervals and the t-test describe sampling variation only. They do not address bias from non-random programme placement or from missing follow-up data, which are discussed in Section 12.

5. Data cleaning and quality assurance

Part A of the assignment asks how inconsistently coded missing values were identified and handled, what quality checks were run, and what the cleaning approach was. This section answers all three. The principle throughout is to retain every record, record each decision in a formula and flag rather than delete.

Sample flow

From 837 surveyed clients to 610 analysed businesses

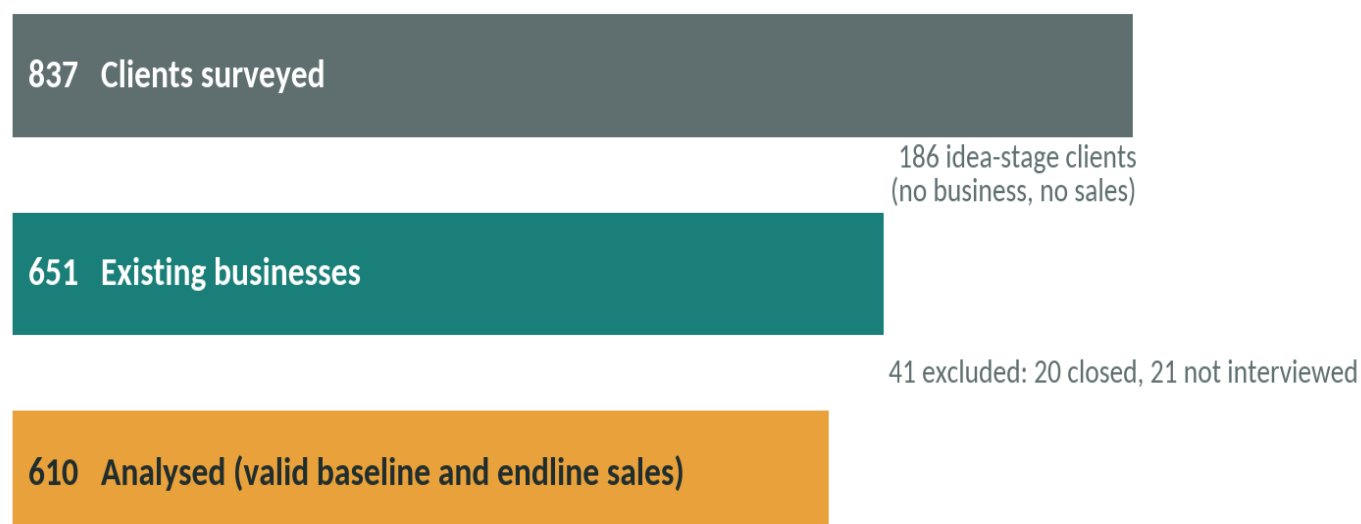


Figure 2. Sample flow from surveyed clients to the analysis sample. Source: *Missing_Code_Audit sheet*.

Of 837 clients, 186 are idea-stage clients with no business yet and therefore no baseline or endline sales; all are in the Training-only tier. Of the 651 existing businesses, 20 had closed and 21 were not interviewed at endline, leaving 610 with valid sales at both time points. **Notably, all 41 lost businesses were Training-only clients:** 41 of 194 existing Training clients (21%), compared with none lost among the 296 Training & advisory and 161 investment clients. This matters for interpretation and is returned to in Section 12.

Identifying inconsistently coded missing values

Each of the 28 variables was audited for the codes 998 and 999, the text "N/A" and blank cells, and for numbers stored as text. In total 5,377 cells are missing or coded missing. The audit shows that the codes are not interchangeable: they reflect different reasons for missingness, which determines how each should be handled.

Table 3. Missing-value codes found in the raw data (*Missing_Code_Audit sheet*)

Code	Cells	Where it appears	Handling rule
999	1,798	Clients not reached at endline (227 on each endline variable) and 206 idea-stage clients on "business operating"	Set to blank; never averaged
998	2	Client age	Set to blank; client kept for all other analysis
"N/A"	186	Business type of idea-stage clients	Set to blank (structural: no business yet)
Blank	3,391	Skip logic (734 with no employees, 379 non-savers) and 227 clients not reached; 3 further blanks in age	No employees = 0; savings read among savers; otherwise blank

Two further patterns deserve mention. First, 832 of the 837 ages were stored as text rather than numbers and would have been silently dropped by many tools; they were converted with VALUE(). Second, the three age entries coded 999 and two coded 998 show that the same variable can use both codes, so the rule must test for each code in every variable rather than assume one.

Cleaning rules applied

Table 4. *Cleaning rules (Parameters and Clean_Data sheets)*

Issue	Rule	Result
Missing codes	IF value is blank, 998, 999 or "N/A" then blank	One consistent representation of missing
Age stored as text	VALUE() conversion; ages outside 15 to 100 set to blank	832 ages converted; 5 set to blank
Impossible employee count	Employees above 50 per MSE set to blank	One value of 120,626 removed
Implausible salary	Monthly salary below 500 CFA set to blank	One value of 1 CFA removed
Skip-logic blanks (employees)	No employees = 0 in e_total_clean	Employee averages use valid answers
Skip-logic blanks (savings)	Amount blank for non-savers; Savings_All_CFA adds 0 for non-savers	Savings analysed among savers and across all households
Variable standardisation	Strata, location and nationality text cleaned	Consistent group labels
New variables	Youth_Status, Refugee_Status, Location_Type, net change, % change, flags	Used in all analysis tables
Record retention	No rows deleted; In_Sales_Sample flag selects analysis rows	837 rows kept; 610 analysed

Quality checks before analysis

Twenty-two checks were run and are listed in Appendix B. Twelve passed outright and ten flagged an issue that was resolved or documented. The checks cover five areas:

- **Uniqueness:** no duplicate client IDs and no duplicated baseline and endline sales pairs.
- **Ranges and plausibility:** ages, employee counts, salaries, sales values and the 0 to 10 satisfaction scale.
- **Logic and consistency:** idea-stage clients report no sales; non-savers report no amounts; every saver reports an amount; cohort and location nest perfectly.
- **Outliers:** five very large net sales gains and two savings amounts above 71,000 CFA were flagged.
- **Sample accounting:** a reconciliation from 837 records to the 610 analysed, by reason for exclusion.

Outliers and sensitivity

Five net gains lie beyond the far-out fence. All five are investment-tier businesses that started with 70,000 to 110,000 CFA in monthly sales and gained between 40,164 and 55,822 CFA. They are plausible large-business results rather than errors, so they were retained. Excluding them lowers the overall mean gain from 9,011 to 8,713 CFA and the investment-tier mean from 13,907 to 12,910 CFA, but relative growth stays near 33% (33.0% overall), so no conclusion changes. Two savers above 71,000 CFA (maximum 320,000 CFA) raise mean savings from 12,212 to 13,043 CFA, which is why the median is quoted as the typical value.

Cleaned dataset delivered

Inkomoko_Cleaned_Dataset.xlsx contains all 837 records with cleaned variables, derived variables and flags, plus a Cleaning_Notes sheet. The workbook version contains the formulas behind each cleaned column.

6. Sales outcomes

Overall progress

Among 610 existing businesses with valid sales at both time points, mean monthly sales rose by 9,011 CFA, from 26,792 to 35,803 CFA, a gain of 33.6%. The 95% confidence interval for the mean gain is 8,379 to 9,642 CFA. The distribution is right-skewed: the median gain is 6,918 CFA, one in four businesses gained less than 3,384 CFA and one in four gained more than 12,578 CFA. Only six businesses (1%) sold less at endline, and the largest decline was 14,433 CFA.

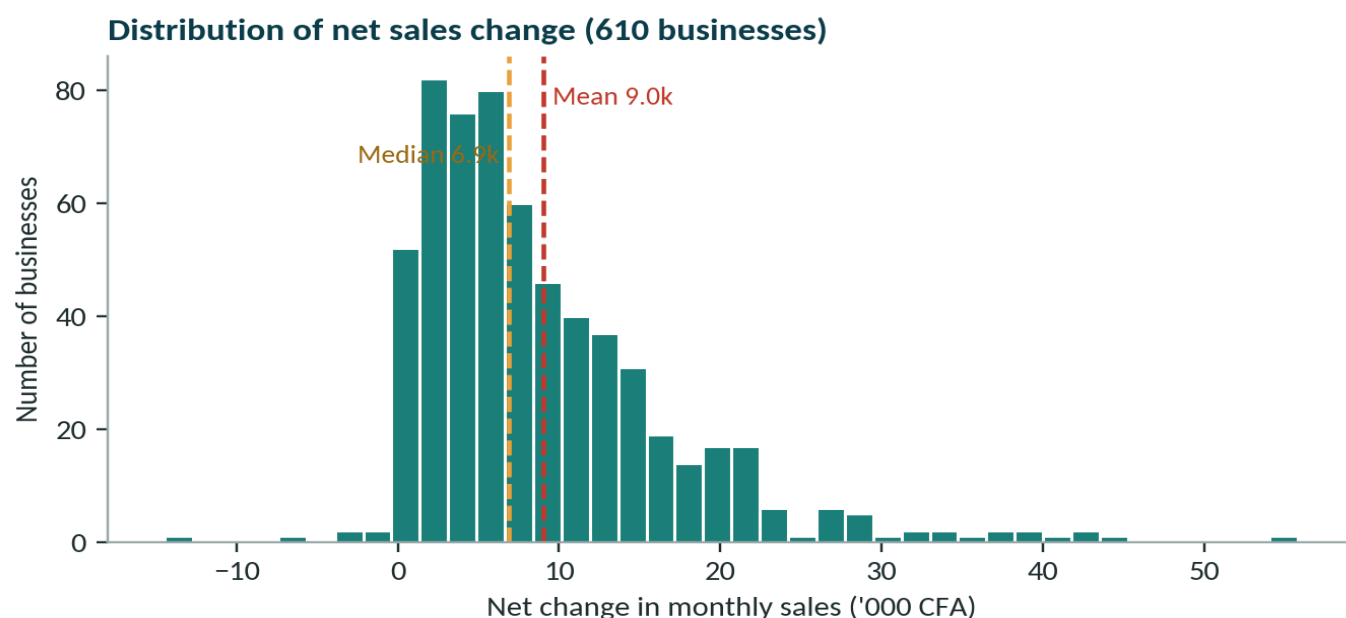


Figure 3. Distribution of net change in monthly sales. Source: Clean_Data sheet (Net_Sales_Change_CFA).

Subgroup comparison

Table 5. Net sales change by subgroup, CFA per month (Sales_Analysis sheet)

Group	n	Mean baseline	Mean endline	Mean net change	% growth	% higher	95% CI of net change
All analysed clients	610	26,792	35,803	9,011	33.6%	99%	8,379 to 9,642
Female	394	26,488	35,382	8,894	33.6%	99%	8,122 to 9,666
Male	216	27,347	36,570	9,223	33.7%	99%	8,129 to 10,317
Refugee	261	26,948	35,831	8,883	33.0%	100%	7,878 to 9,888
Host / non-refugee	349	26,676	35,782	9,106	34.1%	98%	8,297 to 9,914
Hekima Town	214	26,100	34,661	8,561	32.8%	97%	7,566 to 9,556
Lala Camp	189	15,598	21,290	5,692	36.5%	100%	4,784 to 6,600
Sawa Camp	207	37,729	50,235	12,505	33.1%	100%	11,363 to 13,648
Youth (35 or under)	287	23,258	31,189	7,930	34.1%	99%	7,086 to 8,775
Non-youth (over 35)	321	29,818	39,816	9,998	33.5%	99%	9,079 to 10,917
Training	153	19,706	26,096	6,390	32.4%	99%	5,542 to 7,239
Training & advisory	296	22,380	30,081	7,701	34.4%	99%	6,995 to 8,408
Training, advisory, & investment	161	41,639	55,547	13,907	33.4%	99%	12,297 to 15,518

Youth/non-youth excludes two analysed clients with missing age. Intervals that do not overlap indicate real differences: Sawa Camp versus Lala Camp, Training versus the investment tier, and Youth versus Non-youth (marginally). Gender and refugee-status intervals overlap almost completely. Gender and refugee-status intervals overlap almost completely.

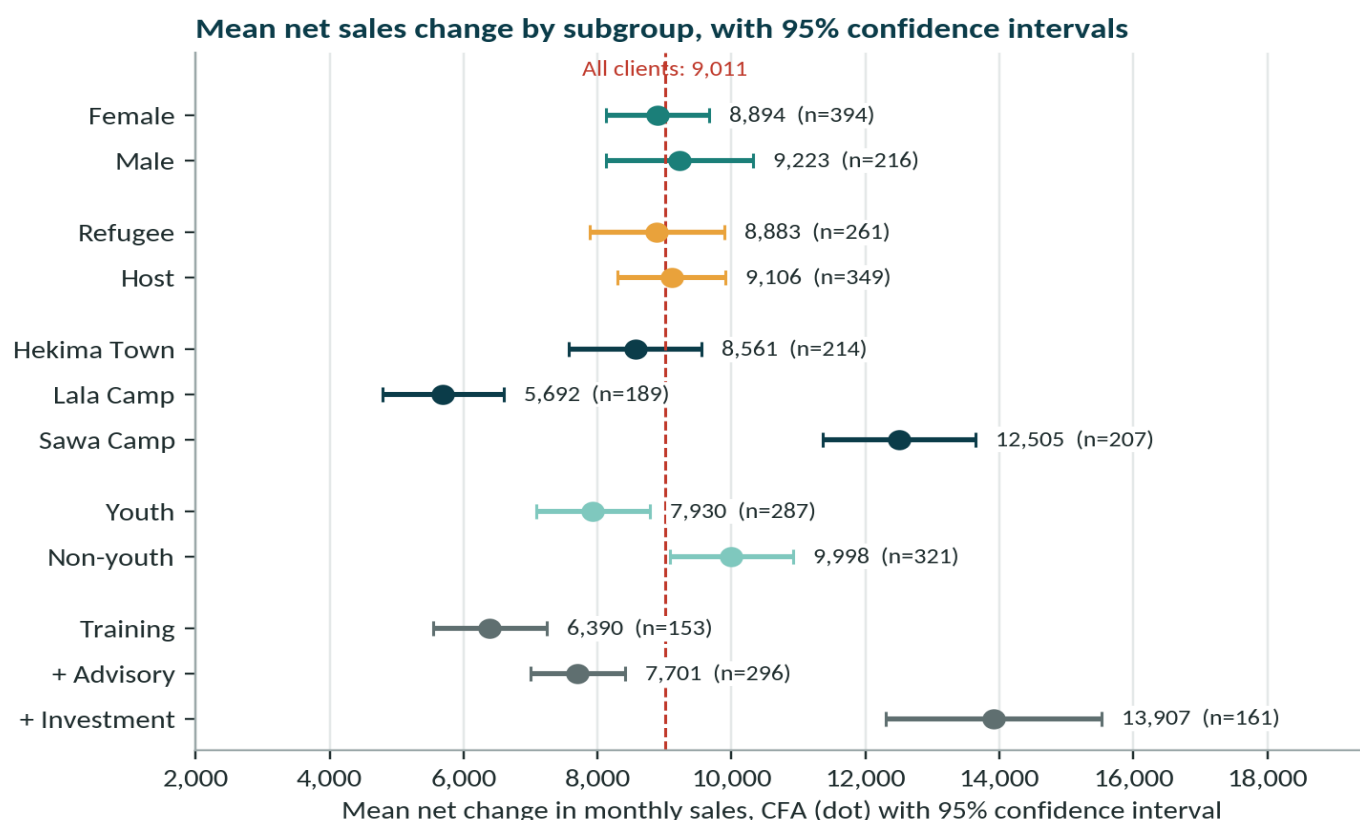


Figure 4. Mean net sales change with 95% confidence intervals. Source: Sales_Analysis sheet.

Absolute gains versus relative growth

The most important interpretive point in the sales results is that the CFA gaps between groups are mostly gaps in starting size. Relative growth lies between 32.4% and 36.5% in all twelve subgroups. Lala Camp, which has the smallest CFA gain, actually has the highest relative growth (36.5%).

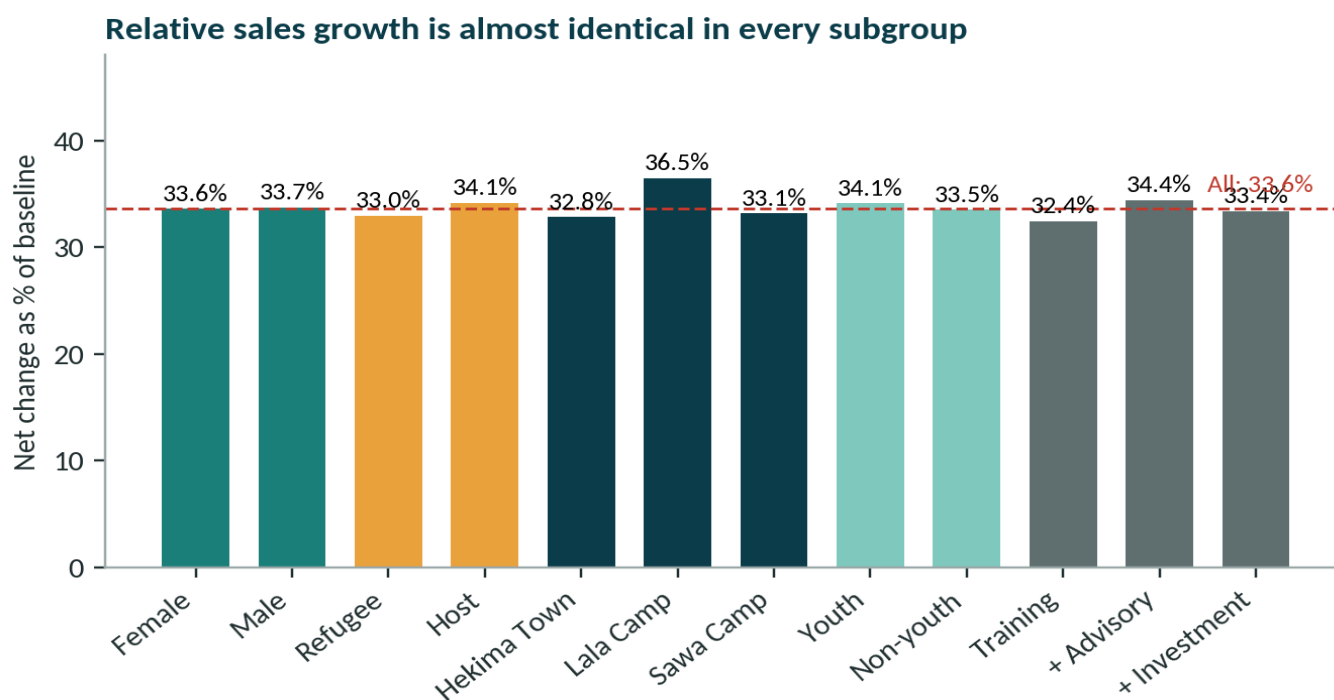


Figure 5. Relative sales growth by subgroup. Source: Sales_Analysis sheet (column G).

Grouping businesses by baseline sales confirms this at the business level. The smallest quarter of businesses (average baseline 9,121 CFA) gained 3,203 CFA, the largest quarter (average baseline 53,269 CFA) gained 17,091 CFA, but median growth was 36% for the smallest and 31% for the largest. The correlation between baseline sales and net change is 0.70, while the correlation between baseline sales and percentage growth is -0.05.

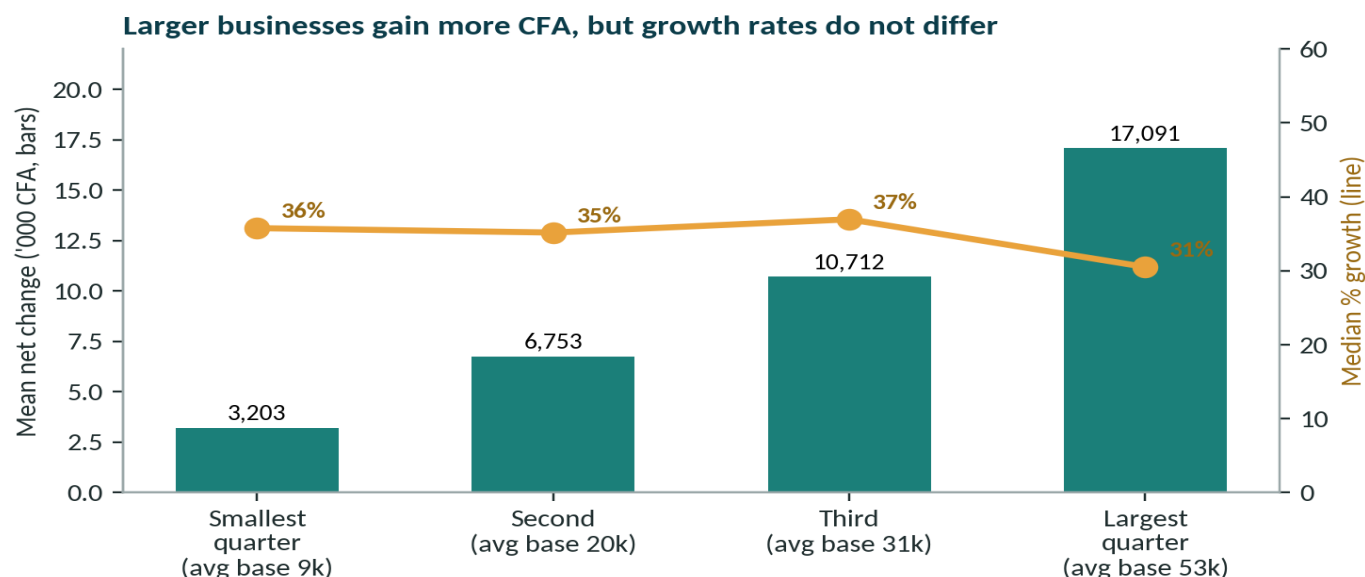


Figure 6. Net sales change and median growth by baseline-sales quarter. Source: Clean_Data sheet.

Support tier: who received what, and what changed

Net gains rise with the depth of support: 6,390 CFA for Training only, 7,701 CFA for Training & advisory and 13,907 CFA for Training, advisory, & investment. The pattern holds in every location. However, the tiers served different businesses. Investment clients began with mean sales of 41,639 CFA against 19,706 CFA for Training-only and 22,380 CFA for Training & advisory clients, and the same ratio appears within each location. Because relative growth is virtually identical (32.4%, 34.4%, 33.4%), the larger CFA gains of the investment tier largely reflect larger businesses, not faster growth.

Mean net sales change (CFA) by support tier and location

	Hekima Town	Lala Camp	Sawa Camp
Training	5,933 (n=55; base 19k)	3,790 (n=54; base 11k)	10,153 (n=44; base 31k)
+ Advisory	7,903 (n=105; base 22k)	4,150 (n=83; base 12k)	10,235 (n=108; base 30k)
+ Investment	12,516 (n=54; base 40k)	10,129 (n=52; base 26k)	18,846 (n=55; base 58k)

Figure 7. Mean net sales change by support tier and location, with sample size and baseline sales. Source: Crosstabs sheets A and H.

Within locations the tier gradient is clear but the interpretation remains the same: investment clients in Lala Camp (baseline 25,596 CFA) gained 10,129 CFA, a 39.6% rise, while Training-only clients in Lala Camp (baseline 10,799 CFA) gained 3,790 CFA, a 35.1% rise. Advisory alone adds little over training in Lala Camp (4,150 versus 3,790 CFA).

Location and refugee status

Location explains more of the variation than any other dimension. Mean gains are 8,561 CFA in Hekima Town, 5,692 CFA in Lala Camp and 12,505 CFA in Sawa Camp, matching baseline sales of 26,100, 15,598 and 37,729 CFA. The refugee versus host comparison (8,883 versus 9,106 CFA) hides a strong interaction with location: refugees in Lala Camp (n = 114, baseline 10,788 CFA) gained only 3,887 CFA, whereas hosts in Lala Camp gained 8,435 CFA and refugees in Sawa Camp gained 12,758 CFA. Hekima Town has no refugee clients.

Mean net sales change (CFA) by refugee status and location

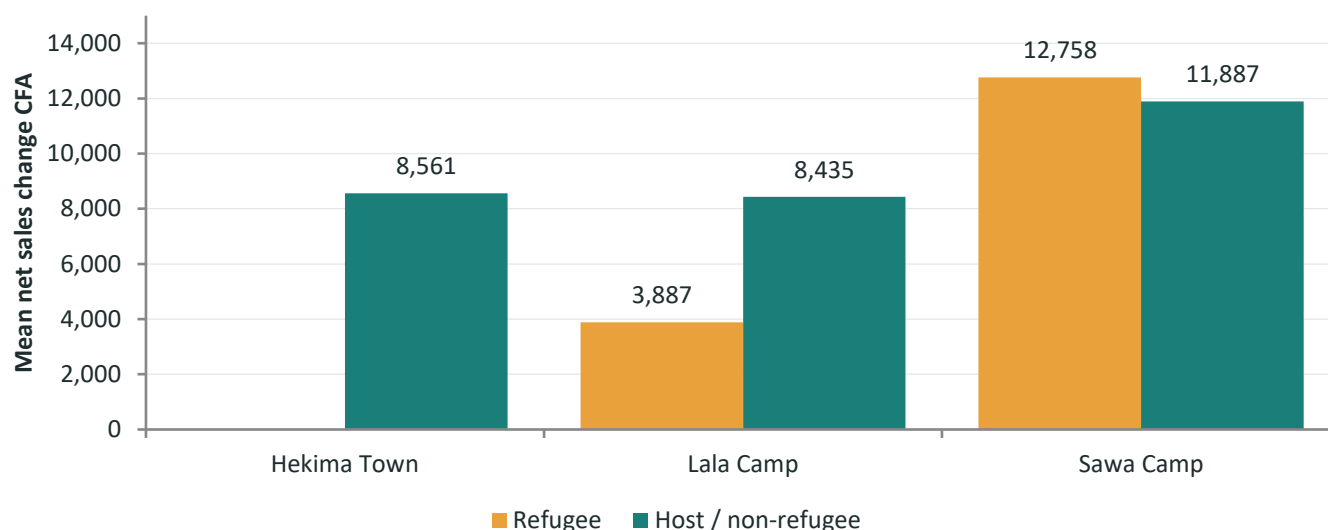


Figure 8. Mean net sales change by refugee status and location. Source: Crosstabs sheet C.

Hekima Town contains all six businesses that declined (2.8% of its 214 analysed businesses), with decreases of 1,500 to 14,433 CFA. No business declined in either camp. The numbers are small but the concentration is worth monitoring.

Gender and youth

Women (65% of the analysis sample) gained 8,894 CFA and men 9,223 CFA. Within each location the gap is small: 8,649 versus 8,381 CFA in Hekima Town, 5,453 versus 6,127 CFA in Lala Camp and 12,450 versus 12,595 CFA in Sawa Camp. Women are therefore not lagging once location is accounted for.

Youth gained 7,930 CFA against 9,998 CFA for non-youth, and youth businesses start smaller (23,258 versus 29,818 CFA), with similar relative growth (34.1% versus 33.5%). Part of the youth gap is geographic: 117 of 287 youth (41%) are in Lala Camp compared with 72 of 321 non-youth (22%). Within every location and tier non-youth still gain somewhat more in CFA terms, so smaller business scale among youth remains a real feature.

Table 6. Mean net sales change (CFA) by youth status, location and intervention (Crosstabs sheets D and E)

Group	Hekima	Lala	Sawa	Training	Advisory Invest.
Youth	7,930	4,660	12,001	5,666	7,117 12,047
Non-youth	8,949	7,369	12,971	7,123	8,257 15,424

Interpretation

Progress in sales is broad-based and statistically strong. Differences between groups are overwhelmingly differences in business scale, and location is the main carrier of that scale. Where the data suggests a genuine gap rather than a scale effect is for refugees in Lala Camp, whose absolute gains are the lowest and whose business base, access to funds and market position are the weakest.

7. Household savings

Household savings are measured by liv_save_hwmuch_CFA, collected only at endline. The analysis therefore compares savings levels across groups; it cannot show that savings have grown. All 610 clients with sales data answered the savings question.

Overall

Three in four households (458 of 610, 75.1%) report saving. Among savers the mean balance is 13,043 CFA and the median 10,000 CFA, with the middle half of savers holding between 3,000 and 20,000 CFA. Counting non-savers as zero, mean household savings are 9,793 CFA. A saver's mean balance equals about 36% of mean endline monthly sales, or roughly 2,175 CFA per household member given an average household size of six.

Savings by subgroup

Table 7. Household savings by subgroup, CFA (Savings_Analysis sheet)

Group	n	Savers	% saving	Mean (savers)	Mean (all)	Mean excl. outliers	Savers' mean / endline sales
All analysed clients	610	458	75.1%	13,043	9,793	12,212	36.4%
Female	394	287	72.8%	12,379	9,017	12,125	35.0%
Male	216	171	79.2%	14,158	11,208	12,359	38.7%
Refugee	261	228	87.4%	12,732	11,122	12,732	35.5%
Host / non-refugee	349	230	65.9%	13,352	8,799	11,693	37.3%
Hekima Town	214	135	63.1%	13,944	8,797	13,414	40.2%
Lala Camp	189	158	83.6%	9,034	7,552	7,053	42.4%
Sawa Camp	207	165	79.7%	16,145	12,869	16,145	32.1%
Youth (35 or under)	287	217	75.6%	11,423	8,637	11,423	36.6%
Non-youth(over 35)	321	239	74.5%	14,393	10,717	12,806	36.1%
Training	153	116	75.8%	11,161	8,462	11,161	42.8%
Training & advisory	296	223	75.3%	13,720	10,336	12,011	45.6%
Training, advisory, & investment	161	119	73.9%	13,609	10,059	13,609	24.5%

Who saves

Saving is more common among refugees (87%) than hosts (66%), and among men (79%) than women (73%). It is almost identical across youth and non-youth (76% and 75%) and across support tiers (76%, 75%, 74%). Location interacts with refugee status: 93% of refugees in Sawa Camp save but only 48% of hosts there, while in Lala Camp 81% of refugees and 88% of hosts save. The lowest rates are hosts in Sawa Camp (48%) and women in Hekima Town (58%, compared with 73% for men there).

Saving rates by location

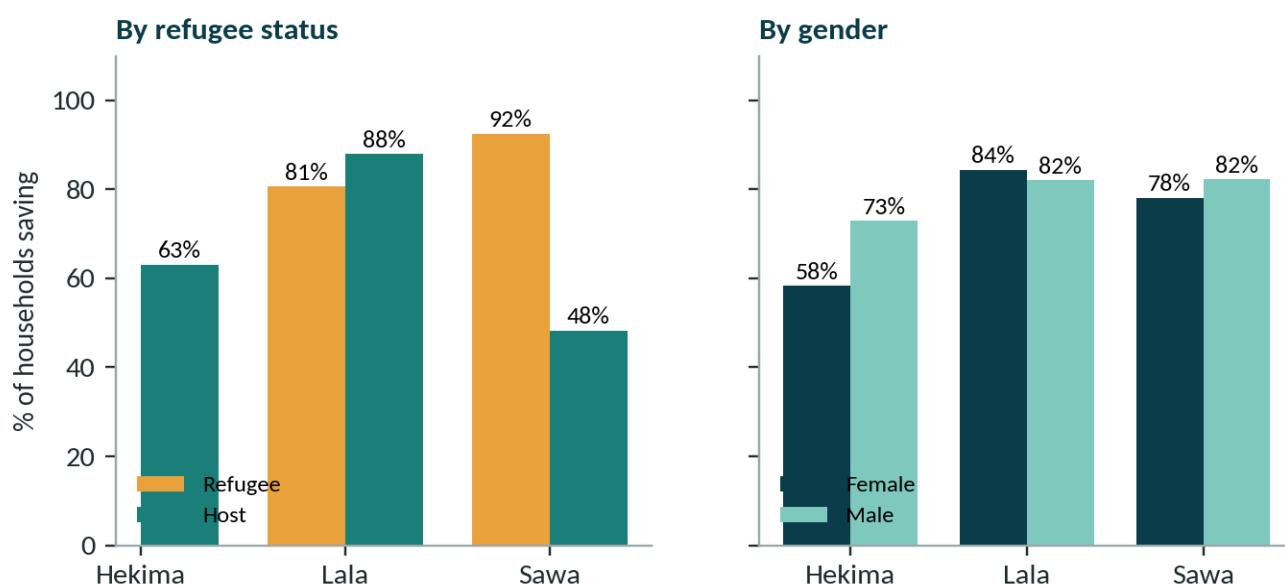


Figure 9. Share of households saving by location, refugee status and gender. Source: Crosstabs sheets F and G.

How much they save

Savings amounts vary by location more than by any other dimension. Savers in Sawa Camp hold 16,145 CFA, in Hekima Town 13,944 CFA and in Lala Camp 9,034 CFA, 44% less than Sawa. In relative terms, however, Lala savers hold 42% of their mean monthly sales against 32% in Sawa Camp, so the shallow balances in Lala reflect small business scale rather than a weaker saving effort. Differences by tier are modest: 11,161 CFA for Training, 13,720 CFA for Training & advisory and 13,609 CFA for the investment tier.

Where households save

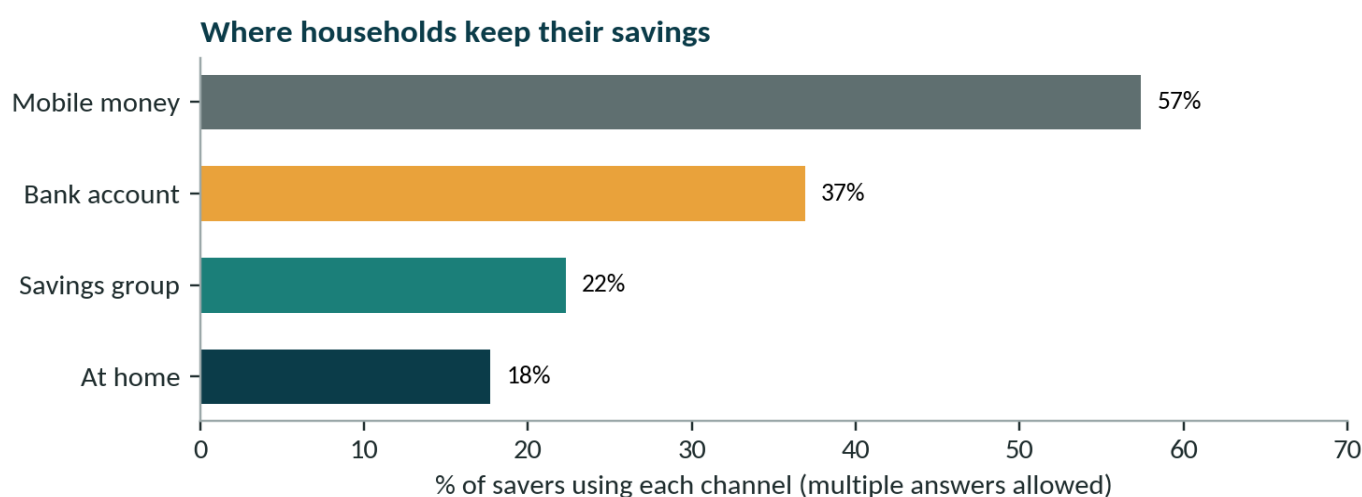


Figure 10. Savings channels used by savers (multiple answers allowed). Source: Clean_Data sheet (liv_save_where).

Mobile money is the most common channel (57% of savers), followed by bank accounts (37%), savings groups (22%) and keeping money at home (18%). This matters for programme design because mobile-money and group-based savings products can reach households that lack bank access, particularly in Sawa Camp where only 38% of respondents have a bank account.

Savings and sales

Among savers, the correlation between the amount saved and the net gain in sales is 0.30. This modest positive relationship is consistent with the theory of change, in that businesses that grew more also hold more savings, but it does not establish causation and may partly reflect business size.

8. Business capability and access to finance

The survey also records endline indicators of business capability and access to finance. They help to explain why the three locations differ and where support could be focused. These indicators exist at endline only, so they describe the current position rather than change.

Business capability and access indicators by location (endline)

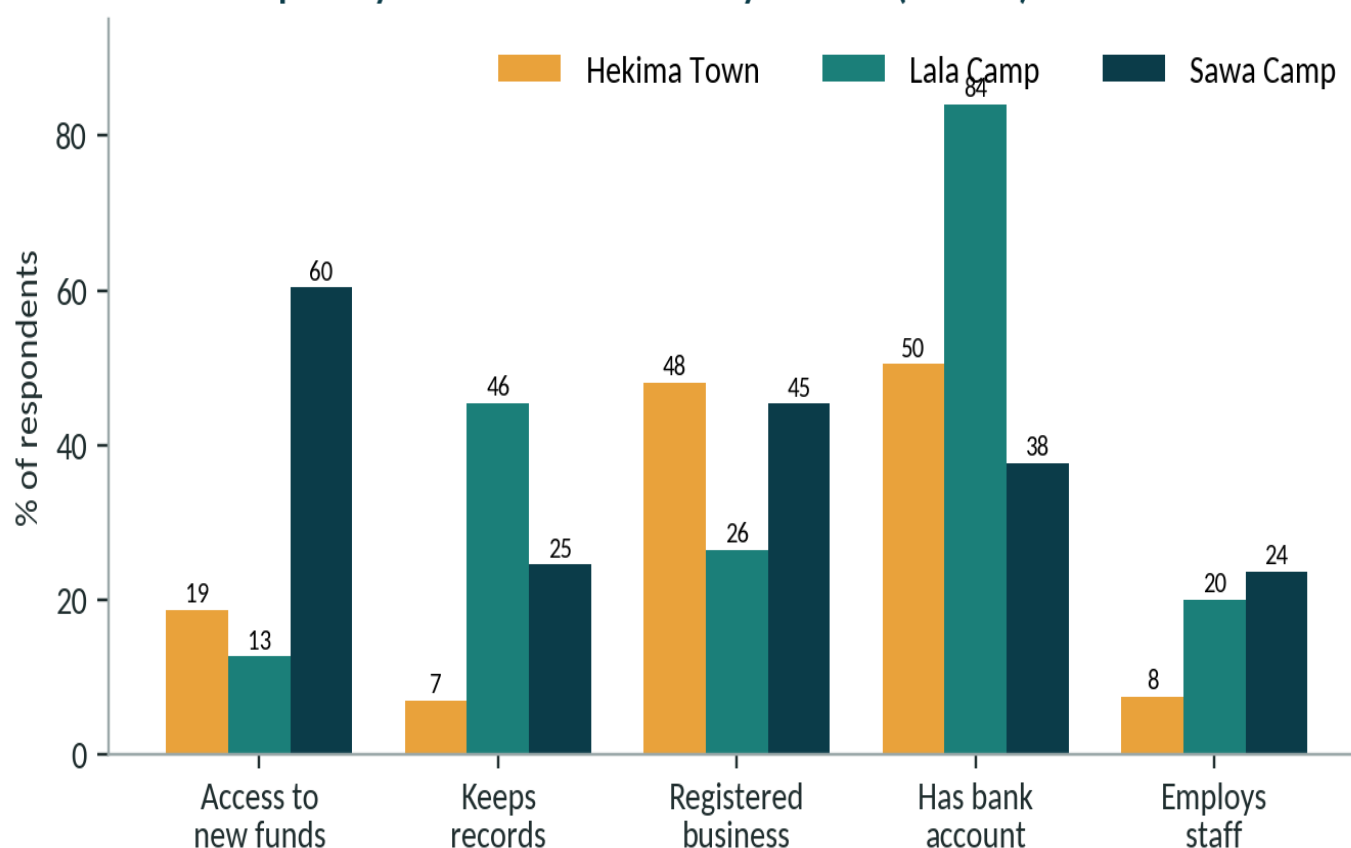


Figure 11. Business capability and access indicators by location. Source: Clean_Data sheet.

Hekima Town is distinctive for its low formalisation of business management: only 7% keep records and 7% employ staff, even though bank account holding (51%) and registration (48%) are comparatively high. Satisfaction with Inkomoko is also lowest here (6.5 of 10 versus 7.2 and 7.1). This is the same location where all six declining businesses appear.

Lala Camp shows high bank account holding (84%) and the highest rate of record-keeping (46%), but the lowest access to new funds (13%), the lowest registration rate (26%) and the smallest business base. For Lala, capital and market access rather than basic business skills appear to be the binding constraints.

Sawa Camp has by far the best access to new funds (60%) and the most employers (24%) but the lowest bank account rate (38%), pointing to informal or mobile-money financial relationships.

Across the sample, 17% of businesses employ staff (271 employees reported, an average of 2.7 per employer), 25% keep records, 57% hold a bank account, 41% are registered and 31% report access to new funds. These low rates show substantial headroom for the advisory component.

9. Implications for the programmes team

This section answers the assignment questions: what is going well, what is not going well, which improvements are needed, and for whom. The scorecard at the end summarizes differences across the five dimensions.

What is going well

- **Near-universal growth.** 99% of businesses sold more at endline and relative growth is about a third in every subgroup, so the core training-led model is working across very different contexts.
- **Equity of outcomes.** Women (65% of analyzed clients) and refugees (43%) grow as fast as men and hosts, and gender gaps within each location are small.
- **Savings habit.** 75% of households save, with refugees leading at 87%. Mobile money is the dominant channel.
- **Client satisfaction.** Satisfaction averages 7.0 of 10 and is highest in the full-package tier (7.2).
- **Data quality.** Duplicates and logic errors are absent, and the data can support detailed subgroup analysis.

What is not going well

- **Lala Camp refugees.** Their gain of 3,887 CFA is the lowest of any group and less than a third of the gain for refugees in Sawa Camp. Lala also has the weakest access to new funds (13%).
- **Hekima Town.** All six declining businesses, the lowest satisfaction, and very low record-keeping and employment.
- **Saving gaps.** Hosts in Sawa Camp (48%) and women in Hekima Town (58%) save least.
- **Tracking and attrition.** 227 clients (27%) have no sales data. All 41 lost existing businesses were Training-only clients, a 21% loss in that tier.
- **Formalization and jobs.** Only 17% of businesses employ staff and 25% keep records.

Improvements needed

- Tailor Lala Camp support towards market access, working capital and savings groups, and test whether the investment tier can be reached by more Lala businesses.
- Add record-keeping, costing and hiring modules to the advisory component, starting in Hekima Town.
- Introduce savings products and group-based saving for Sawa Camp host clients and Hekima Town women.
- Evaluate the added value of advisory and investment services with a comparison group before scaling.
- Improve follow-up of idea-stage clients and closed businesses and extend the same survey instruments to all tiers.

Scorecard by dimension

Table 8. Subgroup scorecard: for whom is the programme working?

Dimension	Sales (net change, growth)	Savings	Assessment
Gender	Female 8,894(33.6%); Male 9,223 (33.7%)	Female 73% saving, 12,379 CFA; Male 79%, 14,158 CFA	On track; watch Hekima women savings
Refugee status	Refugee 8,883(33.0%); Host 9,106(34.1%)	Refugee 87%; Host 66%	On track; Lala refugees and Sawa hosts need attention
Location	Hekima 8,561; Lala 5,692; Sawa 12,505	Hekima 63%, Lala 84%, Sawa 80%; Lala savers hold least	Lala Camp gap; Hekima capability gap
Youth status	Youth 7,930; Non-youth 9,998 (growth 34.1% vs 33.5%)	Youth 76%, 11,423 CFA; Non-youth 75%, 14,393 CFA	Smaller scale among youth; growth equal
Intervention	Training 6,390; Advisory 7,701; Investment 13,907 (growth about 33% in all)	76%, 75%, 74% saving	Cannot yet prove added value of advisory or investment

10. Using the insights to improve programmes

The insights are turned into practical actions below. Each action cites the evidence behind it and a suggested indicator to track. Leads and timings are suggestions for discussion with the programmes team.

Table 9. *Suggested action plan*

Action	Evidence	Suggested lead	Timing	Indicator to track
1. Lala Camp growth package	Lowest baseline (15,598 CFA); refugee gain 3,887 CFA; access to new funds 13% (Sawa 60%)	Programmes team, Lala field office	Next intake	Access to funds; CFA gain for Lala refugees
2. Business-management uplift in Hekima Town	Records 7%; employers 7%; satisfaction 6.5; 6 of 6 decliners	Advisory team	Next 6 months	Record-keeping %; No. of declining businesses
3. Targeted savings products	Sawa hosts 48% and Hekima women 58% saving	Programmes and partners	Next 6 months	% saving in these groups
4. Evaluate advisory and investment value	Growth about 33% in all tiers; investment clients started twice as large	MEL team	Next cohort	Growth difference vs comparison group
5. Reduce attrition	41 of 194 Training clients lost (21%); 186 idea-stage clients without outcomes	MEL and field teams	Next survey round	Follow-up rate by tier
6. Extend measurement	No baseline savings or employment; no consumption data	MEL team	Before next baseline	Coverage of resilience indicators
7. Hiring and formalisation module	Only 17% employ staff; 41% registered	Advisory team	Next 12 months	Jobs created per business
8. Routine MEL dashboard	Location by tier results already built in the workbook	MEL team	Quarterly	Dashboard reviewed each cycle

A learning agenda

- **Does advisory add growth beyond training?** Compare matched Training and Training & advisory clients, or randomise the order of advisory roll-out.
- **Does investment accelerate growth or follow it?** Investment clients may be selected because they were already larger and growing. A phased roll-out among eligible clients would reveal whether capital changes the growth rate.
- **What explains Lala Camp?** Combine the survey with qualitative interviews on markets, customer base and finance in Lala.
- **Is growth sustained?** Add a follow-up round to check that sales gains persist and whether households convert them into savings and consumption.

11. Alignment with the theory of change

The pathway to test is: business support (training, advisory and business loans) → expanded businesses → higher sales and savings → household resilience, ultimately moving clients out of poverty. The table assesses each link against the evidence from this survey.

Table 10. Pathway assessment

Pathway stage	Evidence from the endline survey	Status	Confidence
1. Business support	837 clients reached (380 Training, 296 Training & advisory, 161 investment); satisfaction averages 7.0 of 10; 186 are idea-stage clients with no business yet	Delivered	High
2. Expanded businesses	Mean monthly sales up 33.6%; 17% employ staff (271 jobs reported); 25% keep records; 57% bank; 41% registered	Partly evidenced	Medium: endline only for jobs and formalisation
3. Higher sales and savings	+9,011 CFA a month; 99% of businesses grew; 75% save; savings correlate 0.30 with sales gain	Strong	Medium: no comparison group; no baseline savings
4. Household resilience	Savings about 2,175 CFA per member; no consumption, asset or poverty measures	Not yet shown	Low: not measured

Table 11. Pathway indicators by support tier (TOC_Indicators sheet)

Indicator	All	Training	Training and advisory	Training, advisory, and investment
Clients served	837	380	296	161
Existing businesses with sales data	610	153	296	161
Mean baseline monthly sales (CFA)	26,792	19,706	22,380	41,639
Mean endline monthly sales (CFA)	35,803	26,096	30,081	55,547
Mean net change (CFA)	9,011	6,390	7,701	13,907
Relative growth	33.6%	32.4%	34.4%	33.4%
% of businesses with higher sales	99.0%	98.7%	99.3%	98.8%
% with at least one employee	16.9%	17.0%	13.9%	22.4%
% keeping business records	24.9%	27.5%	23.3%	25.5%
% with a bank account	56.6%	62.1%	55.4%	53.4%
% with access to new funds	31.0%	32.0%	28.0%	35.4%
% of households saving	75.1%	75.8%	75.3%	73.9%
Mean savings of savers (CFA)	13,043	11,161	13,720	13,609
Savings per household member (CFA)	2,175	1,860	2,230	2,384
Mean satisfaction (0 to 10)	7.0	6.9	6.9	7.2

Assessment

The results show clear progress along the first three links of the pathway. Support has been delivered, businesses are larger by sales, and households are saving. This progress is consistent across gender, refugee status, youth status and tier.

Two qualifications apply. First, the pathway from support to higher sales is shown as a before-and-after change, not as an effect attributable to Inkomoko, because there is no comparison group. Second, relative growth is the same in every tier, so the data supports the proposition that Inkomoko clients grow, but not the more specific proposition that deeper support increases growth. The step to household resilience cannot be assessed: the survey has no consumption, asset or poverty-status indicators and no baseline savings, so even the savings result is a level, not a change.

What would complete the pathway

Add a household consumption or poverty-likelihood module and baseline savings; track employment changes; follow clients over time; and introduce a comparison group so that gains can be attributed to Inkomoko rather than to inflation, seasonality or survival.

12. Limitations and caveats

This report is descriptive. It documents what changed for supported entrepreneurs and for whom, but it was not designed to prove why. The limitations below are grouped by the kind of inference they constrain, and each ends with what the data can and cannot support.

Attribution and design

- **No comparison group.** The design is before-and-after. Gains may reflect inflation, seasonality, general economic conditions or maturation, not only Inkomoko support. The reported gains therefore describe change among supported clients; they are not an estimate of programme impact.
- **Non-random tier assignment.** Investment clients were already larger, so tier differences in CFA gains cannot be seen as programme effects. The larger gains in the advisory and investment tiers may reflect who entered those tiers rather than what the tiers delivered.

Sample and data coverage

- **Survivorship and attrition.** Idea-stage clients and closed or untraced businesses are excluded from the sales analysis. All 41 lost existing businesses were Training-only clients, so the Training tier result may be overstated if lost businesses were weaker. Zero attrition in the advisory and investment tiers is also unusual and may reflect how the sample was drawn. The 99% share of businesses that grew applies to surviving, traceable businesses, not to every client enrolled.
- **Small cells.** Some two-way cells (for example Training-only clients in Sawa Camp, $n = 44$) are modest; cells below 30 are flagged in the workbook. Differences between small cells should be treated as indicative rather than conclusive.
- **Savings at endline only.** No baseline exists, so savings cannot show progress; they are compared as levels. The finding that 75% of households save; describes the position at endline, and the correlation of 0.30 between savings and net sales change is an association among savers with no established direction of cause.

Measurement scope and interpretation

- **Unmeasured resilience outcomes.** Household consumption, assets and poverty status are not in the dataset. The final links of the theory of change are therefore untested, not disproved.
- **Currency and prices.** All amounts are nominal CFA with no inflation adjustment. If prices rose over the period, real growth is lower than the 33.6% reported, although comparisons between groups are less affected because all clients were measured over the same period.
- **Imaginary dataset.** The data is described as an imaginary 2024 endline survey, so findings illustrate the analytical approach and should not be read as evidence about real Inkomoko performance.

What remains robust

Taken together, these limitations narrow the claims the report can make without overturning its findings. The direction and breadth of change (sales higher in 604 of 610 businesses, three in four households saving) and the near-identical growth rates across groups are facts about this sample and hold whatever the study design. Causal attribution to Inkomoko, the added value of the advisory and investment tiers, and household resilience cannot be claimed from this data, and the report does not claim them. The measurement upgrades set out in Section 10 would remove each of these constraints.

13. Conclusion

Overall assessment

Inkomoko-supported entrepreneurs in this endline survey show strong, broad-based progress. Monthly sales rose by a third on average, nearly every business grew, and three in four households save. The analysis finds no evidence that women, refugees or youth are being left behind in relative terms; their smaller CFA gains reflect smaller starting businesses, and growth rates are almost identical across all groups.

In figures, mean monthly sales rose from 26,792 to 35,803 CFA, a net gain of 9,011 CFA per business (95% confidence interval 8,379 to 9,642 CFA, or +33.6%), and 604 of 610 businesses grew. The result is statistically unambiguous (paired $t = 27.98$, $p < 0.001$) across every subgroup examined. Alongside it, 75% of households (458 of 610) save.

Where the gaps lie

The most important gaps are structural rather than demographic. Lala Camp, and refugees there in particular, combine the smallest business base with the weakest access to finance. Hekima Town combines low business formalisation with the only declining businesses and the lowest satisfaction. Sawa Camp hosts and Hekima women save least. The advisory and investment tiers go with larger absolute gains, but the data cannot yet show that they raise growth.

Sawa Camp businesses gained 12,505 CFA against 5,692 in Lala Camp, alongside access to funds of 60% against 13%. Absolute gains rise by support tier (6,390 to 13,907 CFA) while relative growth stays near 32 to 34%, so scale, not momentum, separates segments. In Hekima Town only 7% of businesses keep records and 7% employ staff, satisfaction is lowest at 6.5 out of 10, and all six businesses whose sales fell are located there. Refugees save more than hosts (87% against 66%), but rates fall to 48% among hosts in Sawa Camp and 58% among women in Hekima Town.

What this mean for the theory of change

The evidence supports the first half of the pathway. Training, advisory services and finance are associated with larger businesses, higher sales and widespread household saving. It cannot yet speak to the second half. Employment, household consumption, assets and poverty status were not measured, and savings have no baseline, so the step from business growth to household resilience and exit from poverty is untested rather than refuted.

Priorities for action

The recommendations in Section 10 follow directly from this evidence, and with them Inkomoko can test the full theory of change, from business support to household resilience.

- **Close the capital and scale gap in Lala Camp** through market access, working capital and savings groups.
- **Prove the value of the advisory and investment tiers** with a phased roll-out or matched comparison before scaling.
- **Strengthen business management in Hekima Town**, where record-keeping, staffing and satisfaction are weakest.
- **Target savings products** at hosts in Sawa Camp and women in Hekima Town, who save least.
- **Upgrade measurement** by adding baseline savings, employment tracking and a household consumption module.

Overall conclusion

Inkomoko clients are growing and saving, and the progress is broad-based. The question is no longer whether clients grow, but whether Inkomoko support is the reason, whether the advisory and investment tiers add value beyond business size, and whether the gains reach household resilience. Closing the Lala Camp capital gap, strengthening Hekima Town and upgrading measurement will turn a strong descriptive result into evidence of impact.

Appendix A: Missing-value audit

Counts for each raw variable from the Missing_Code_Audit sheet. Variables with no missing or coded values are listed in the note below the table.

Table 12. Variables with missing or coded values (837 records)

Variable	998	999	N/A	Blank	Total	Likely meaning
cd_age	2	3	0	0	5	Data-entry error; stored as text
bd_biztype	0	0	186	0	186	Idea stage: no business yet
bd_biz_operating	0	206	0	0	206	Idea stage / not asked
cd_sole_income_earner	0	227	0	0	227	Not interviewed at endline
cd_household_size	0	0	0	227	227	Not interviewed at endline
bd_biz_registered	0	227	0	0	227	Not interviewed at endline
e_do_you_have_employees	0	227	0	0	227	Not interviewed at endline
e_total	0	0	0	734	734	Skip logic (no employees)
e_fulltime_employe_salary_CFA	0	0	0	734	734	Skip logic (no employees)
sr_have_bank_account	0	227	0	0	227	Not interviewed at endline
sr_access_newfunds	0	227	0	0	227	Not interviewed at endline
sr_no_customers_monthly	0	0	0	227	227	Not interviewed at endline
sr_records_keep	0	227	0	0	227	Not interviewed at endline
liv_saving	0	227	0	0	227	Not interviewed at endline
liv_save_hwmuch_CFA	0	0	0	379	379	Skip logic (non-savers)
liv_save_where	0	0	0	379	379	Skip logic (non-savers)
mkl_where_sell_prod_servc	0	0	0	227	227	Not interviewed at endline
sales_baseline_monthly_CFA	0	0	0	227	227	Idea stage / no sales data
sales_endline_monthly_CFA	0	0	0	227	227	Idea stage, closed or untraced
satisf_scale	0	0	0	30	30	No rating given
Total across all variables	2	1,798	186	3,391	5,377	

No missing or coded values were found in ID, nationality, strata, gender, business status, intervention, location or cohort.

Table 13. Why sales data are missing (837 clients)

Status	Clients	Share
Analysed	610	72.9%
Idea stage: no baseline business	186	22.2%
Business not operating	20	2.4%
Existing business: no sales data	21	2.5%

Appendix B: Data-quality checks

All checks are live formulas on the DQ_Checks sheet. PASS means the result equals the expected value; REVIEW means an issue was found and the action in the last column was applied.

Table 14. Twenty-two data-quality checks (12 passed, 10 flagged and resolved)

Check	Result	Status	Action
Records in raw file	837	PASS	All retained
Duplicate client IDs	0	PASS	None
Duplicate sales pairs (analysed rows)	0	PASS	None
Ages stored as text	832	REVIEW	Converted with VALUE()
Ages coded 998 or 999	5	REVIEW	Set to blank
Cleaned ages outside 15 to 100	0	PASS	None
Employee count above plausible maximum	1	REVIEW	Value 120,626 set to blank
Salary below plausible minimum	1	REVIEW	Value 1 CFA set to blank
Employers with no valid employee count	1	REVIEW	Retained as missing
Idea-stage clients reporting sales	0	PASS	Logic consistent
Zero or negative sales values	0	PASS	None
Clients whose sales declined	6	REVIEW	Plausible; retained and reported
Net-change outliers (3 × IQR)	5	REVIEW	Investment-tier large businesses; retained
Savings outliers above fence	2	REVIEW	Retained; sensitivity shown
Savers with no amount	0	PASS	None
Non-savers with a savings amount	0	PASS	Skip logic consistent
Satisfaction scores outside 0 to 10	0	PASS	None
Cohort-location mapping breaks	0	PASS	Perfectly nested
Existing businesses not operating	20	REVIEW	Excluded; reported as attrition
Idea-stage clients outside Training tier	0	PASS	All Training-only
Analysed clients missing age	2	REVIEW	Excluded from youth comparison only
Size of net-sales analysis sample	610	PASS	As expected

Appendix C: Variable definitions and derived variables

Table 15. *Derived variables in the cleaned dataset*

Variable	Definition
Refugee_Status	Refugee if camp-based refugee; otherwise Host / non-refugee
Youth_Status	Youth if cd_age_clean is at most 35; Non-youth if over 35; blank if age missing
Location_Type	Refugee camp for Lala and Sawa; Town for Hekima
In_Sales_Sample	1 if both baseline and endline sales are valid, otherwise 0
Net_Sales_Change_CFA	Endline minus baseline monthly sales
Pct_Sales_Change	Net change divided by baseline sales
Sales_Increased_Flag	1 if net change is greater than 0
Savings_All_CFA	Savings amount with non-savers set to 0
Saver_Flag	1 if the household saves, 0 if not, blank if unanswered
Savings_Outlier_Flag	1 if savings exceed $Q3 + 3 \times IQR$ (71,000 CFA)
NetChange_Outlier_Flag	1 if net change lies outside the $3 \times IQR$ fences
Sales_Sample_Status	Analyzed; Idea stage; Business not operating; or Existing business with no sales data
Duplicate_ID_Flag	1 if the client ID occurs more than once
Raw_Missing_or_Coded_Cells	Count of 998, 999 and blank cells in the raw record

Appendix D: Workbook guide and sources

Table 16. *Sheets in Inkomoko_MEL_Analysis_Workbook.xlsx*

Sheet	Part	Contents
README		Purpose, sheet guide, cleaning approach, definitions, colour legend
Summary	B	Headline indicators, sample flow, paired t-test and confidence interval
Dashboard	B	Four native charts linked to the analysis tables
Sales_Analysis	B	Net sales change by five dimensions with n, means, % growth and confidence intervals
Savings_Analysis	B	Savings by the same dimensions; outlier sensitivity; distribution statistics
Crosstabs	B, C	Two-way splits of sales and saving; baseline by tier and location
TOC_Indicators	C	Pathway indicators by support tier
DQ_Checks	A	22 live data-quality checks
Missing_Code_Audit	A	Counts of 998, 999, N/A and blanks per variable
Clean_Data	A	Cleaned dataset, formula-driven from Raw_Data
Raw_Data	A	Original data as received
Parameters	A, B	Editable thresholds and assumptions

How to audit any number

Pick a cell on Sales_Analysis, for example the mean net change for Lala Camp. Its formula is an AVERAGEIFS over the Net_Sales_Change_CFA column of Clean_Data with the location and sample-flag criteria. Net_Sales_Change_CFA in turn is an IF formula on the cleaned endline and baseline columns, each of which is a formula on Raw_Data using the thresholds on Parameters. Nothing is typed in as a constant except the editable parameters.

Software and sources

- Software: Microsoft Excel formulas (COUNTIFS, AVERAGEIFS, SUMIFS, SUMPRODUCT, QUARTILE, MEDIAN, CORREL, TDIST); recalculated and error-checked with LibreOffice with zero formula errors across 36,785 formulas.
- Data: Inkomoko MEL test brief and sample endline dataset, September 2026.
- Method: Tukey, J. W. (1977). Exploratory Data Analysis. Addison-Wesley (outlier fences). Standard paired t-test and normal-approximation confidence-interval formulae.
- Charts in this report were produced from workbook values with Python (matplotlib); figures match the workbook.